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Peltoniemi, Mirva; Sihvonen, Antti Tapani

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Craft and Industrialization: Uncovering the Plasticity and Frontier of Craft

Mirva Peltoniemi[a] **and Antti Sihvonen**[b]

[a]*Tampere University;* [b]*University of Helsinki*

ABSTRACT Craft as an approach to work has been proposed as an alternative to industrial mass production. While industrialization has led to the replacement of craft logic with industrial logic to create uniformity and scale operations, the role of craft in industrialization remains ambiguous. Building on the analysis of engineering work performed at an R&D laboratory in the pulp and paper industry, we explore how and why craft enables industrialization. Our findings reveal how elements of craft manifest in engineering work by helping to transform ideas into useful objects, balancing scientific accuracy with practical usefulness, and fostering a sense of community and identity within the engineering field. These findings point to *the plasticity of craft logic*, suggesting that elements of craft can be transferred into new contexts and reinterpreted to address new objectives. Considering that these activities took place in an R&D laboratory that drove industrialization, they also reveal the existence of a *craft frontier* that escapes standardization by providing a context for crafting candidates for future standards. Craft frontier enables the coexistence of craft and industrial logics, where craft can contribute to industrialization by establishing foundations for managerial control without being absorbed into it.

Keywords: craft, craft frontier, engineering, industrialization, plasticity

INTRODUCTION

Craft is an approach to work that prioritizes human skills, dedication, and communality (Kroezen et al., 2021; Ravasi et al., 2025) and it constitutes an institutional logic that organizes social reality and guides behaviour (Kroezen and Heugens, 2019; Thornton and Ocasio, 2008). Since craft logic reflects pre-industrial and artisanal ways of production, it is at odds with industrial logic, which prioritizes standardization, profit, and rational

Address for reprints: Mirva Peltoniemi, Unit of Industrial Engineering and Management, Faculty of Management and Business, Tampere University, P.O. Box 553, FI-33014, Finland (mirva.peltoniemi@tuni.fi).

Order of authorship is alphabetical. Authors contributed equally to this work.

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organization structure (Kroezen and Heugens, 2019). As industrialization has progressed, craft logic has been overtaken by industrial logic as traditional crafts have transformed into modern industries (Blundel and Tregear, 2006; Thornton, 2002), where pockets of craft have survived within otherwise mass production-oriented sectors (Berg, 2014; Bettiol and Micelli, 2014; Bonfanti et al., 2018). Recently, craft logic has resurged as a post-industrial alternative to mass manufacturing by responding to distinct consumer needs (Beverland, 2005; Cabras and Bamforth, 2016; Kroezen and Heugens, 2019).

Despite the juxtaposition between craft logic and industrial logic, several scholars have argued that craft skills are needed to produce the preconditions of mass production (Dormer, 1997; Holt and Popp, 2016; Pye, 1995). It has also been suggested that during industrialization the craftsperson's workshop transformed from a place of production to a 'way station to the factory' (Sennett, 2008, p. 84). Although these studies shed light on the role of craft in industrialization, they have not fully theorized how craftwork integrates into industrialization or how craft logic might shape industrialization. Consequently, our understanding of the relationship between craft and industrialization remains incomplete.

To dig deeper into the dynamics between craft and industrialization, we turn to engineering because of its craft origins (Braverman, 1974; Torstendahl, 2021) and its contribution to industrialization and the standardization of work (Mokyr, 1992; Mowery and Rosenberg, 1999). To understand how craftwork integrates into industrialization, we focus on how elements of craft manifest in engineering work that creates machines and systems of mass production. To guide our study, we ask the following research question: *How and why does craft enable industrialization*?

We adopted an interpretive historical approach (Vaara and Lamberg, 2016) and analysed engineering work at Central Laboratory, the joint R&D laboratory of Finnish pulp and paper companies from 1916 to 2009. We concentrated on the period from 1920 to 1972 because during that time the industry scaled up to mass production. Our analysis covered 56 engineering projects focused on developing methods and equipment for characterizing different pulps and paper grades. Guided by sensitizing concepts from craft theory (Bowen, 2006), we applied grounded theory techniques to analyse work in these projects (Gioia et al., 2013). This approach enabled us to elaborate craft theory in the context of engineering by identifying how different elements of craft manifest and interact in engineering projects to influence work outcomes.

We make two primary contributions by studying the role of craft in enabling industrialization. First, our findings show how craftwork facilitated the industrialization of Finnish pulp and paper production through the engineering work performed at Central Laboratory. In this work, elements of craft are visible in how Central Laboratory engineers transformed ideas into useful objects, balanced scientific accuracy and practical usefulness, and built a community and identity for themselves. We theorize that the presence of craft elements in engineering work stems from the *plasticity of craft logic*, which enables the logic to be transferred into new contexts and reinterpreted to address new goals (Notteboom et al., 2013; Strambach, 2010). This was the case in the Finnish pulp and paper industry, where craftwork originating in individual pulp and paper mills was reshaped into *preparatory workmanship* (Pye, 1995, pp. 131–132) that enabled developing the preconditions of mass production at Central Laboratory. Therefore, rather than

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being marginalized, craft logic can be moulded to contribute to industrialization and subsequently to the creation of industrial logic.

Second, we develop the concept of the *craft frontier* to explain how craftwork can shape industrialization. We define the craft frontier as an epistemic boundary at the limits of industrialization where craftwork is used to develop new candidates for future standards. Because the outcomes of such development work are inherently uncertain, craft skills and attitudes are required to steer the work through which machines, tools, and methods of industrialization are created (Dormer, 1997; Pye, 1995). Inside the frontier, work is organized according to industrial logic, while beyond it lie non-systematic and ad hoc ways of working, as well as natural phenomena that are not yet sufficiently understood to enable the development of standards. In the Finnish pulp and paper industry, work on the craft frontier was concentrated at Central Laboratory where engineers developed candidates for future standards that would be applied at the pulp and paper mills. Therefore, the craft frontier can both sustain and eliminate craft from an industry: actors operating at the frontier retain discretion to hone their skills and determine what constitutes good work, while others focus on applying what is developed at the frontier. This division allows craft and industrial logics to co-exist and the engineers following different logics to work toward the mutual goal of industrialization (see Reay and Hinings, 2009).

THEORETICAL BACKGROUND

Craft as an Approach to Work

Craft has several meanings, including artisanal production, categorical, and narrative construction, and a distinct approach to work and organizing (Ravasi et al., 2025). Our study falls under the third category since we conceptualize craft as an institutional logic which is composed of material practices, values, and attitudes that organize social reality and guide behaviour (Thornton and Ocasio, 2008). As a logic, craft constitutes a timeless and humanistic approach to work that empowers workers to decide how their work is performed (Kroezen et al., 2021). This discretion is guided by the pursuit of quality (Beverland, 2005; Fox Miller, 2017; Sennett, 2008) and the goal of producing useful and functional things (Becker, 1978; Holt and Yamauchi, 2019).

Human skills and *attitudes* are the defining elements of craft that distinguish it from work done or controlled by machines (Kroezen et al., 2021). Craft involves a high degree of skill developed over time through learning by doing and trial-and-error and the production of high-quality outcomes for the sake of doing good work (Becker, 1978; Bozkurt and Lara Cohen, 2019; Sennett, 2008). Craft encompasses self-directed work rooted in embodied, technical, and aesthetic mastery (Bell et al., 2021), and much of this work is not conducive to standardization (Orr, 2016, pp. 13–15). Based on these characteristics, we can distinguish *workmanship of certainty*, which is associated with industrial production where the outcome is predetermined, from the *workmanship of risk*, where the result depends on the skills and judgment of the craftsperson (Pye, 1995, p. 20). Workmanship of risk can be further divided into productive and preparatory workmanship, where *productive workmanship* is focused on creating goods for sale while *preparatory workmanship* focuses

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on making the 'plant, tools, jigs, and other apparatus which make the workmanship of certainty possible' (Pye, 1995, p. 52). Thus, craft skills can be used for a variety of purposes.

Craft attitudes orient workers toward dedication to their craft (Cattani et al., 2017) and establish craft communities where knowledge is shared and relationships are organized based on skill (Barley, 1996; Holt and Yamauchi, 2019). They also involve an inclination to tinker and explore new ideas and ways of working to develop one's skills and advance the craft more broadly (Fox Miller, 2017; Holt and Popp, 2016). Following this ethos, Hodson (2010) describes the craft mandate as being based on peer training and peer supervision, resulting in high levels of effort, pride, and work satisfaction. Therefore, craft attitudes provide a way to organize relationships based on different levels of mastery and develop a community around a shared interest in advancing the craft.

Craft and the History of Industrialization

The mainstream narrative of the history of industrialization views craft as the original method of production, which has been replaced by machine-controlled, large-scale manufacturing processes that have led to the deskilling of workers (Blundel and Tregear, 2006; Braverman, 1974; Wallace and Kalleberg, 1982). For Braverman (1974, p. 155), the scientific and engineering revolution associated with industrialization displaced craftwork through the application of increasingly scientific knowledge of natural laws, which was used to create an increasingly fragmented subdivision of labour preferred by capitalist management.

During the history of industrialization, the craft logic that foregrounds skills, reputation, and community to organize work was gradually replaced by industrial logic that emphasizes formal organization structure, standardization, and market performance to control work (Kroezen et al., 2021; Kroezen and Heugens, 2019; Rennstam and Paulsson, 2025). This change in the dominant logic enabled increasing the scale of operations (Berg, 2014) and creating consistent output (Blundel and Tregear, 2006). It also resulted in the separation of intellectual work from the work of execution (Braverman, 1974, pp. 78–79).

In the narrative of industrialization, craft has a place in the margins. It remains a part of rural and remote cultures of making (Berg, 2014; Gasparin and Neyland, 2022), or it is used to tailor mass-produced goods to fit customer needs (Bettiol and Micelli, 2014) and handle service activities that cannot be industrialized (Holmes, 2015). However, several studies have also challenged this narrative by arguing that individuals such as Josiah Wedgwood applied craft skills to advance industrialization (Holt and Popp, 2016) and by noting that mass production requires *preparatory workmanship* to create the machines of mass production (Pye, 1995, p. 52). In this line of thinking, craft is not understood purely as handcraft but as a way of working that enables the craftsperson to take charge of technology and how it is used (Dormer, 1997, p. 102). During industrialization, movements such as distributism also tried to reconcile craft and industrial logics by advocating the dispersion of wealth and work to small community-based organizations where skilled manual work would shape how manufacturing machines were used (Holt, 2026).

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A substantial share of research on craft within organization studies has focused on the return of craft in post-industrial societies, where it is seen as a fundamental element of working and organizing (Ravasi et al., 2025). Many organizational scholars have recognized the persistence of craft within industrialized manufacturing (Bell and Vachhani, 2020; Blundel and Tregear, 2006; Form, 1987; Holt and Popp, 2016). Bell and Vachhani (2020) argue that craft is a relational condition that can exist in industrial settings and that mechanized practices are not the antithesis of craft. Industrialization has also created demand for new skills (Form, 1987; Holt and Popp, 2016) since artisanship can facilitate industrial design (Bettiol and Micelli, 2014), and artisanal practices can be combined with industrial ones (Blundel and Tregear, 2006). Thus, craft has a role in many industrialized sectors by servicing particular consumer needs or fulfilling tasks that cannot be automated.

Engineers and Industrialization

Industrialization relies on the work of engineers (Mokyr, 1992; Mowery and Rosenberg, 1999). The machines, production processes, and control mechanisms developed by engineers have been vital in making industrialization possible (Braverman, 1974). Engineers have also had a key role in the development of the modern industrial enterprise (Chandler, 1977). Although the sociology of professions emphasizes the role of formal education, engineering also embodies craft elements as it originates from both craft guilds and universities (Krause, 1996) and because it integrates craft and science to create useful things (Barley and Orr, 2018).

As a profession, engineering seeks to apply science to develop systems and standards that benefit manufacturing companies (Weitz and Shenhaav, 2000). In doing so, engineering has been shaped by design, science, and systems thinking (Petroski, 1996). At the core of engineering lies the integration of mathematical and scientific analysis, trial-and-error learning, and contextual knowledge in the pursuit of 'making *things* and making them work' (Barley and Orr, 2018, p. 30).

There is variation in how much discretion engineers have in their work. The division between elite and rank-and-file engineers (Layton, 1971, p. 79; Meiksins, 1988) indicates that some get to focus on honing their skills and developing new things while others focus on running operations. In this regard, the division between R&D and operations is crucial since standardization has stripped shopfloor experts of their professional discretion in favour of following a standard test manual (Reynolds, 1986). Economic historians have documented how standardization conducted by engineers has contributed substantially to the productivity growth of the 20th century (Spencer and Temple, 2016). Organization studies have also covered the increase in quality that results from engineers adopting standards (Beck and Walgenbach, 2005; Sandholtz, 2012) as well as how these reduce uncertainty by making activities rational and predictable (Weitz and Shenhaav, 2000). Within the modernist movement of the 20th century, standardization was seen as a virtue and a basis of aesthetics (Guillén, 1997).

Overall, prior literature has understood industrialization as a process that gradually replaces craft logic with industrial logic to increase the scale of operations and create uniformity. Much of the recent literature on craft has focused on the role of craft following widespread industrialization. Yet, several studies have recognized that craft skills are needed to produce the tools and machines that make industrialization possible in the first place

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(Dormer, 1997; Holt, 2026; Holt and Popp, 2016; Pye, 1995; Sennett, 2008). However, they have not fully theorized the role of craft in industrialization, especially how craftwork integrates into industrialization and how craft might shape industrialization. Thus, there is a need to develop our understanding of the role of craft in industrialization and how craft contributes to this process. To understand these dynamics, we focus on engineering work because of its craft origins and important role in industrialization. Next, we present Central Laboratory as the context for studying the role of craft in industrialization.

RESEARCH CONTEXT: CENTRAL LABORATORY

In the pulp and paper industry, the production process begins in the forest and concludes at the printing press. To control all the phases in between, it is necessary to measure things. To sell the end products, it is also crucial to characterize them using unambiguous metrics. For these reasons, Central Laboratory's member companies tasked it with developing testing and analysis methods and equipment for use in the industry. This organization was founded in 1916 and disbanded in 2009. The R&D laboratory was jointly funded by Finnish pulp and paper companies that formed cartels to manage the export of pulp, paper, and cardboard. Most Finnish pulp and paper firms were cartel members throughout our study period.

The starting point was a reality in which no established methods existed and the understanding of what to measure was limited. In 1922, a meeting was held at Central Laboratory with 17 engineers from various mills to address the question of creating uniform methods for process control at all mills (Grönvik, 1966, p. 31). This was concluded to be an important task and a 'Committee on Norms' was established with four mill representatives and three Central Laboratory experts. It was a forerunner for the various technical committees that were to guide the work at the lab in the years to come. The first task assigned to Central Laboratory by the committee included developing methods for determining the hardness, bleachability, strength, colour, and physical and chemical purity of sulphite and sulphate pulps.

What Central Laboratory produced was a constantly updated catalogue of standardized methods, prototypes of physical measuring devices, measurement scales relevant for different applications, and a definition of the scope of application of each method. This work also gained international recognition, and the chairmanship of the ISO subcommittee on testing methods for chemical pulping was placed at Central Laboratory in 1958. Many of the subsequent ISO standards were based on the testing methods originally developed at Central Laboratory.

Our study period extends from 1920 to 1972. During this time, the technological paradigm of the Fourdrinier-type paper machine fed with various types of chemical and mechanical pulps stayed the same (Nykänen et al., 2005, pp. 26, 300). Pulp enters through the wet end of the machine, travels on a wire-mesh belt through an architecture of rolls that press and dry it, and paper exits on a roll in a continuous process. The key raw material of Nordic coniferous trees, whose long fibres are especially well-suited for papermaking, also remained the same. However, there was a marked increase in the size of the paper machines: the average width increased from 634 cm in 1910 to 2251 cm in 1974 (Järvinen et al., 2012, p. 43). As the quality expectations for printed goods became more

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demanding, there was also an increased focus on delivering consistently high-quality goods for the export market (Heikkinen, 2000). The technological change, therefore, predominantly focused on deepening while scaling up, and quality improvements took place within the same technological paradigm.

In the Finnish pulp and paper industry, systematic research and development work was mainly assigned to Central Laboratory. This work largely replaced development work previously performed by production site engineers. However, production site engineers participated in the R&D work through technical committees (24 at their height) that guided the planning and selection of projects and hosted test runs at production scale. Each technical committee (e.g., chemical pulping, mechanical pulping, newsprint) had a representative from each mill for which the respective technical area was relevant. By cooperating with the technical committees, Central Laboratory had a crucial role in the development of the Finnish pulp and paper industry.

We originally read about the development of testing and analysis methods and equipment at Central Laboratory from corporate histories (Grönvik, 1966; Levlin, 2010). What piqued our interest was the work that went into determining variables to be measured and coming up with new ways to measure them. For instance, opacity and thickness need to be balanced when newspapers are printed: low opacity results in ink being visible from the other side of the paper, which can be countered by increasing the thickness of the paper. However, this solution increases costs, as heavier paper requires more raw material and is more expensive to transport. Therefore, the relevant measures form an interconnected network where changes in one metric affect the other characteristics. Different pulps and paper grades are specified through different trade-offs between characteristics. Moreover, paper grades are combinations of different types of pulp (mechanical or chemical with various sub-categories), which means that there is a need to understand how pulp characteristics translate into paper characteristics.

METHOD

Our overarching research strategy is interpretive history. Studies in this tradition are 'based on an intensive qualitative examination of historical sources, with a focus on understanding the meaning of the events in question' (Vaara and Lamberg, 2016, p. 642). The interest is on actions, their meanings for the actors involved and their historical context. Interpretive historians ask 'large questions in small places' (Joyner, 1999, p. 1). The small place we focus on is Central Laboratory where engineering work constituted the actions and events that we interpret. By exploring the work through the lens of craft, we add richness to our understanding on how craft enables industrialization.

The reports that we studied provide a window to engineering work because they were written by the engineers as the work was performed and the primary audience was the engineering community. Industrialization as a historical phenomenon represents the aggregate outcome of engineering work, and the concrete result of scaling up to mass production is known. These ingredients contribute to the historical veracity of our study. We approached our empirical setting with a theoretical

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pre-understanding of craft in organization studies, and we allowed categories to emerge from data, subsequently theorizing with the help of the established understanding of craft. These choices contribute to achieving the gold standard of dual integrity (Maclean et al., 2016).

We see the work reported in our data as *exceptional normal actions* (Hargadon and Wadhwani, 2023, p. 690), as it does not represent the work of the average engineer but the work of leading engineers who are a minority in any industry. They, however, appear commonly in industries seeking technological progress. Our interpretive historical approach guided us to select archival episodes that inform us on the work of developing analysis and testing methods and equipment. Further, this approach guided us to interpret the episodes as communications to the engineering community that is interested in 'making *things* and making them work' (Barley and Orr, 2018, p. 30).

Data Collection

Central Laboratory was an agent of the member firms and it had to report on how their financial contributions were used, what results were achieved, and how decisions were made. As a result, there is a vast collection of documentation detailing these matters. We were able to access the reports at the Central Laboratory archive in Espoo, Finland. Since the organization is no longer operational and there was no archivist to work with us, we were free to search through the shelves independently. This freedom reduced concerns about the strategic use of the past to manage perceptions (Suddaby et al., 2010).

We analysed 'Reports' that were sent to the production sites of member firms. Each report describes the content and outcomes of one project. It was necessary for the reports to be very detailed as they had to convince a knowledgeable audience of the best way to do things and to pre-empt questions concerning alternative solutions. Projects were assigned to the lab by technical committees. When a decision was made to go ahead with a project, a report had to be produced detailing the work and its outcomes. Therefore, the reports form a comprehensive universe of the work performed within the joint R&D programme at Central Laboratory, rather than a curated one controlled by archivists (see Popp and Fellman, 2020).

We chose the period from 1920 to 1972 for two reasons. First, it was the golden age of the joint R&D programme, which started in 1920. During the 1960s, some member firms started to question the costs and benefits of the collaboration. This led to a reorganization of the decision-making and funding structures of the joint R&D programme in 1972, and thereafter the activities started to shift toward firm-specific R&D projects. Second, during this time the industry scaled up to mass production. This resulted from systematic efforts to improve the efficiency of the production processes by developing standardized practices.

During the study period, a total of 1119 reports were produced. We went through the titles of the reports and included those that concern measuring something or determining a quality. This revealed 196 reports that were relevant to our study. Other project types included exploring the physical and chemical characteristics of raw materials, quality control surveys, and travel reports.

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Analysis

We applied grounded theory techniques to code our data (Gioia et al., 2013; Strauss and Corbin, 1998). To guide the analysis process, we used craft theorizing to provide us with sensitizing concepts (Bowen, 2006) that directed our attention to different aspects of craft in engineering. Therefore, our use of grounded theory techniques focused on elaborating craft theory in the context of engineering rather than discovering new theory (see Suddaby, 2006 on the different uses of grounded theory). By using these techniques, we examined how different elements of craft manifest in engineering work and how they interact within engineering projects to shape work outcomes.

We began coding by having two authors independently list events and activities from the project reports. In coding the raw data, we used existing theorizing on craft to sensitize us to what to look for (Bowen, 2006). After going through multiple projects, we began to compare the coded events and activities. This helped us pinpoint activities that were similar in nature. We grouped these codes into categories following first-order coding (Gioia et al., 2013), where both the engineers' activities and their own interpretations of those activities were used to develop the codes (Van Maanen, 1979). For instance, many projects depicted scientific principles and how they could be put into practice, which we coded as 'establishing how a (scientific) principle can be executed in practice'. These activities differed from 'constructing new equipment to execute a principle', which was described separately in the reports and served a distinct function. Therefore, these two activities were treated as being distinct from each other. Developing the first-order codes was an iterative process and we sought to develop a consensual understanding on the similarities and differences between the codes when differences in opinion emerged.

As we progressed with the analysis, we used axial coding to search for similarities and differences between the first-order codes to develop more abstract themes (Strauss and Corbin, 1998; Van Maanen, 1979). For instance, we combined codes related to recognizing the differences between laboratory and factory work under the category of 'relating laboratory and shopfloor practice' since they were associated with the distinction between correctness and practical usefulness which is an element of craft (Sennett, 2008). By comparing the emerging themes to extant theory on craft, we formulated theoretically informed second-order themes (Gioia et al., 2013). After forming the initial categories and themes, we continued to refine the coding through selective coding until we reached theoretical saturation (Strauss and Corbin, 1998; Van Maanen, 1979). Simultaneously, we grouped the second-order themes under three aggregate dimensions to develop an overview of how different elements of craft manifest in engineering work at Central Laboratory. Figure 1 presents the data structure of our coding and portrays how we moved from raw data to first-order categories, second-order themes, and aggregate dimensions. In total, our coding covered all 12 departments and included 56 engineering projects with 554 events and activities.

To understand the dynamic relationships between the emergent themes, we analysed how the second-order themes interact during engineering projects. This analytical step focused on reanalysing the event chronologies we had initially coded for each project. We began by comparing how activities related to our themes manifested in individual

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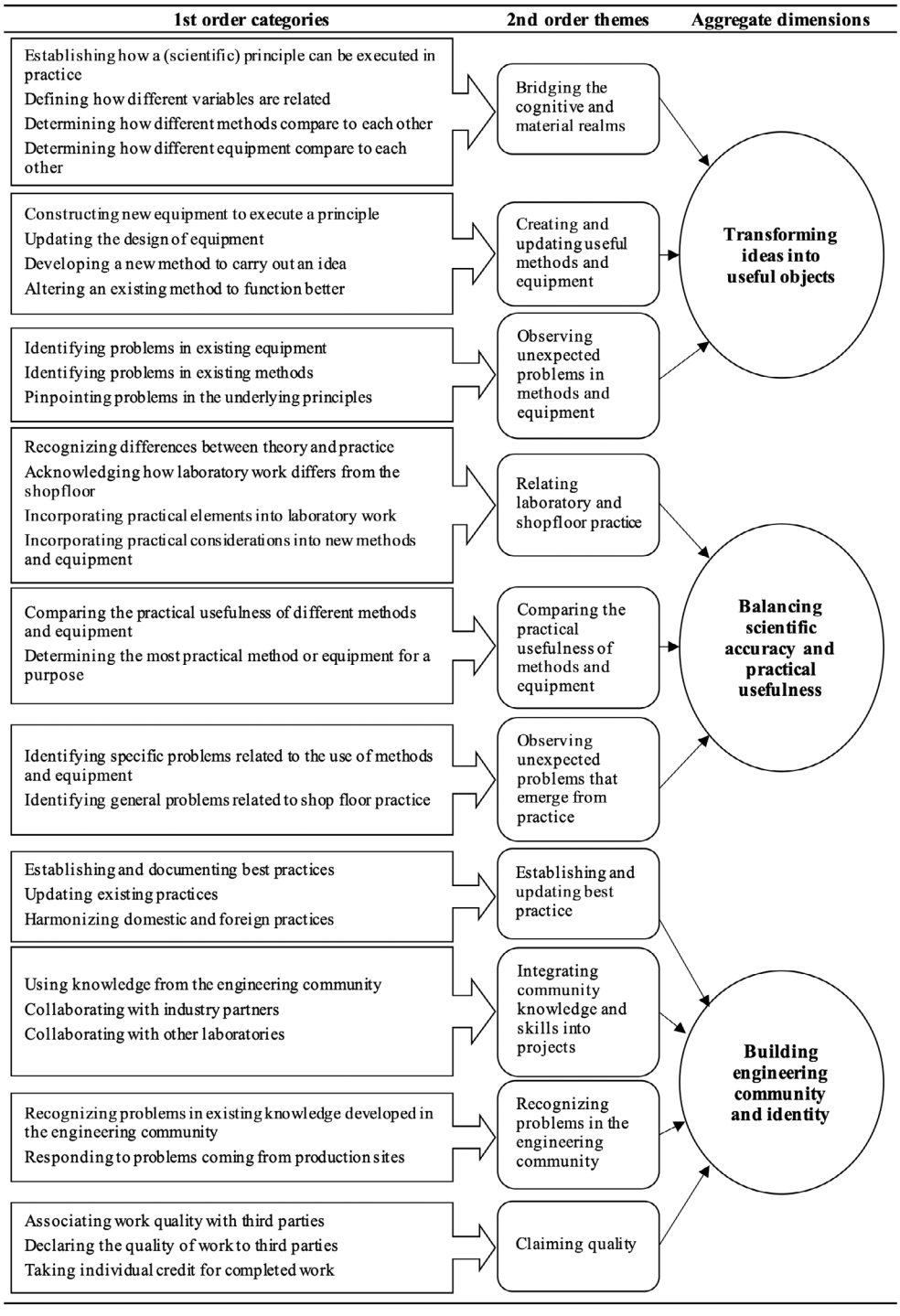


Figure 1. Data structure

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projects. This analysis led to the realization that each aggregate dimension included second-order themes related to both identifying and solving problems. By analysing these activities, we found that projects involved cycles of problem-finding and problem-solving, and the subsequently developed solutions influenced work outcomes as they were embedded in the projects. To summarize how craft influences engineering processes, we used a visual mapping strategy (Langley, 1999) to synthesize how the different elements interact and influence engineering projects (see Figure 3). These findings were then integrated into our final model of how craft elements manifest in engineering work.

While developing a model of how craft manifested in engineering work at Central Laboratory, we sought to understand how these patterns of events and actions link to the increasing scale of operations in the Finnish pulp and paper industry. This led to a realization that while craft was being replaced by standardized work in the pulp and paper mills, the craft logic originating from those mills was being reinterpreted at Central Laboratory to develop the methods and machines that promoted standardization. This recognition led to the discovery of two key dynamics that form the core of our theorization, which pertain to the transformation of craft into being part of the work of industrialization and the recognition of how craft enables the creation of new standardized ways of working. After going back and forth between existing theory on craft and these empirical observations, we theorized them as *plasticity of craft* which conceptualizes the transfer and reinterpretation of craft logic into different contexts, and *craft frontier* as an epistemic boundary that separates the development and application of standardized ways of working and keeps craft relevant in an industry that increasingly follows industrial logics.

FINDINGS

As shown in Figure 2, our model identifies three key dimensions through which elements of craft logic manifest in engineering work at Central Laboratory. These dimensions are: (1) transforming ideas into useful objects; (2) balancing scientific

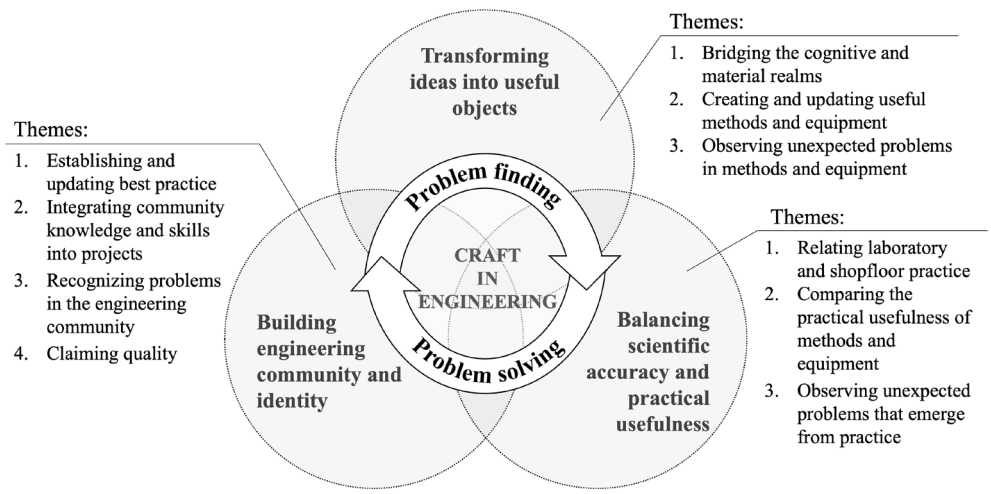


Figure 2. Model of craft in engineering

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accuracy and practical usefulness; and (3) building engineering community and identity. When these dimensions are set in motion during engineering projects, activities in each dimension can help identify and solve problems. In doing so, they contribute to the problem-solving cycles inherent in engineering and influence project outcomes. To understand the model, we first elaborate on the three dimensions that show how elements of craft manifest in engineering work and then examine how they contribute to engineering projects.

Transforming Ideas into Useful Objects

Engineering focuses on the application of science for commercial purposes. At Central Laboratory, this meant transforming ideas and scientific principles into testing methods and equipment that could be used in the pulp and paper industry. The need for systematic development was a key reason for the founding of the laboratory, and in 1942, this idea was succinctly presented in a document outlining analysis and research methods.

> If these kinds of results are to be replicated and compared, the methods need to be precisely defined. For this reason, the industry has several times asked Central Laboratory to make detailed descriptions of their methods and provide copies of these descriptions to mill laboratories and other interested parties. (Analysis and research methods report from 1942)

The development of these methods and equipment reflects the first manifestation of craft logic in engineering work: the transformation of ideas into useful objects. It comprises three themes: (1) bridging the cognitive and material realms; (2) creating and updating useful methods and equipment; and (3) observing unexpected problems in methods and equipment. Next, we elaborate on these themes further, while Table I provides illustrative evidence from our data.

Bridging the cognitive and material realms helped to develop understanding of what an idea or an object meant in different forms and how these forms were related. Bridging the realms meant testing how scientific principles could be executed in practice and analysing how different variables were related. For instance:

> Water, that can be removed from a sheet by pressing, is bonded in the pores between fibres due to the capillary effect. Because water fills the aforementioned pores, it can be removed by decreasing the volume of the pores, i.e., by applying pressure. (Mechanical pulping report 36 from 1957)

Here we see how the theoretical understanding of wood fibre characteristics and the capillary effect relate to the material process of drying new sheets of paper through the application of pressure. Spelling out these relationships and studying them generated new understanding of how to effectively dry paper. In addition to creating new knowledge, bridging the cognitive and material realms also involved developing understanding of how existing methods and equipment compared with each other and executed their underlying principles. These relationships were rarely given and establishing relationships

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Table I. Transforming ideas into useful objects: Themes, categories, and quotations

Second-order themes and first-order categories	*Illustrative quotations*
1. Bridging the cognitive and material realms	
A. Establishing how a (scientific) principle can be executed in practice	Based on the tests concerning the method of making the sheet, it was concluded that in the boiling and bleaching tests the sheets should be made from unground and S-R 35 ground pulp and sheets with weight of 67 and 200 g/m^2 should be made unpressed and with a pressure of 5.4 kg/cm^2. (Cellulose chemistry report 30 from 1957)
B. Defining how different variables are related	The results indicate that there is quite a clear and straightforward correlation between G.B. Brightness, Leukometer, Photovolt and Beckman values. (Cellulose analysis report 21 from 1953)
C. Determining how different methods compare to each other	When it comes to the weight of volume, methods for making the sheets differ considerably from each other: The CL method gives the smallest values whereas the German method gives the largest. (Beating and strength testing report 23 from 1960)
D. Determining how different equipment compare to each other	The goal of this research project has not been to compile conversion charts for new machines but only to form an understanding of their interrelationships. (Cellulose analysis report 7 from 1949)
2. Creating and updating useful methods and equipment	
E. Constructing new equipment to execute a principle	The first machine for keeping the plastic test pieces within the set conditions was designed following the DIN 50907 norm. (Inorganic chemistry report 13 from 1962)
F. Updating the design of equipment	At the moment, a new plotter is under construction for A. Ahlström Plc. Several key improvements have been made to the structure of the electromechanical memory. (Mechanical pulping report 32 from 1955)
G. Developing a new method to carry out an idea	For this purpose, the previously tested method to measure colour absorption in CL was developed to fit its purpose, and the influence of different test conditions on the results was studied. (Cellulose theory report 17 from 1953)
H. Altering an existing method to function better	Therefore, we have had reason to alter the method to suit our conditions. Acetylation is performed without activation and the measurement of results has been simplified so that after acetylation, the measurement of the optic cloudiness of the reaction solution is done immediately after the analysis with a Lumetron-photocolorimeter. (Rayon analysis report 3 from 1948)
3. Observing unexpected problems in methods and equipment	
I. Identifying problems in existing equipment	Sheets dried with the Board department's sheet dryer results in considerably lower breaking and bursting strength and a little higher tearing resistance than for sheets dried by the CL normal method. This likely depends on the sheet drier's high 105 centigrade drying temperature. (Beating and strength testing report 17 from 1955)
J. Identifying problems in existing methods	The performed experiment shows that when determining the alpha number using the Swedish standard method, change in suction power can create up to 1.4 unit changes in the results. Also the density of the glass filter can affect the alpha number, but less than suction power (max. 0.5 units). (Cellulose analysis report 16 from 1952)

(Continues)

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Table I. (Continued)

Second-order themes and first-order categories	*Illustrative quotations*
K. Pinpointing problems in the underlying principles	The assumption is that all fibres have the same weight per length. This assumption is not entirely correct because a fibre's width varies. With cellulose, the error is comparatively small, whereas with easily fibrillating fibres, such as linen and hemp, this assumption cannot be made. (Fibre analysis report 2 from 1958)

required skill and exacting work as can be seen in the following example from a report on methods for testing pulp:

> Calculating from the Enso-number for air-dry pulps one can approximately find the Bergman-number by means of the following formulae, which we have worked out from very extensive observation material. (Description concerning methods of testing wood pulp from 1927)

As illustrated by the previous examples, Central Laboratory engineers pushed the envelope by generating a more precise understanding of the technical preconditions of good work. At the same time, laboratory engineers actively reshaped learning by doing and trial-and-error, long used by craftspeople at the paper mills, by combining these practices with scientific knowledge and a more systematic approach to development work. In doing so, previously mill-specific and ad hoc efforts to enhance papermaking, largely grounded in the craft of local engineers and workers, were transformed into a centralized effort to define good work.

Bridging the cognitive and material realms generated understanding that could be put into practice. This allowed *creating and updating useful methods and equipment*. At Central Laboratory, this work meant the development of methods and equipment for the pulp and paper industry. For instance:

> After a few approximate experiments, two useful methods were developed that will be later entitled: CL method number 1 and CL method number 2. (Mechanical pulping report 8 from 1949)

Here we see how initial experiments prepared the ground for developing two new wood fibre sorting methods. Since these methods differed in how they applied the core principles, their viability was then studied to determine how well they filled their purpose. A similar approach was also used to develop prototypes that allowed testing ideas in practice to learn and refine their designs. In addition, existing methods were also updated when new knowledge was developed:

> Because our previously developed method had plenty of deficiencies, we replaced the previously used solvent with petroleum ether. (Organic chemistry report 23 from 1949)

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This instance relates to the extraction of pine oil from cellulose and how its loss could be minimized through the development of measurement methods. The work of creating and updating methods and equipment represents the most visible aspect of engineering work because the results of this labour could directly influence work practices at the mills.

Part of the work of transforming ideas into useful objects also involved *observing unexpected problems in methods and equipment*, since putting ideas into practice could generate unforeseen issues. Observing unexpected problems helped to identify challenges in existing methods and equipment and pinpoint deficiencies in their underlying principles. For instance, the aforementioned project to develop wood fibre sorting methods was influenced by the identification of shortcomings in how an existing method worked:

> The larger the water volumes used, the faster the sorting takes place. The published tables show, however, that the conclusions are not completely correct. This has been confirmed in experiments performed at Central Laboratory. (Mechanical pulping report 8 from 1949)

The inaccuracies identified in the project presented an opportunity to develop a new method for executing the underlying idea more efficiently. Similarly, 'unsatisfactory results' and 'deficiencies' were often the spark that kindled new projects since they helped to identify problems on the knowledge frontier. One reason for the recognition of deficiencies and new problems was the increasing sophistication of methods and equipment over time which pushed the boundaries of knowledge forward. For instance, a 1953 report on the colour absorption of cellulose noted that earlier tests conducted in 1946 lacked sufficient accuracy to yield conclusions, but advances in technology and research methods had since enabled further investigation. In doing so, unexpected problems and technological development helped to define what kind of problems existed on the frontier of established knowledge and when they could be addressed.

From a theoretical perspective, transforming ideas into useful objects shows how elements of craft logic manifest in the ways engineers engage with materials and objects. Our data reveal how engineers bridged the conceptual and material realms (Barley, 1996; Bell et al., 2019; Bettiol and Micelli, 2014) and explored what different materials and objects could do (Bell and Vachhani, 2020). This work culminated in the creation of methods and equipment for industrial production, which can be understood as *preparatory workmanship* (Pye, 1995, p. 52) that depends on the craftsperson's skill to control the process of making (Dormer, 1997, p. 102). While historical accounts highlight how early development work in the Finnish pulp and paper industry was dispersed across individual paper mills and relied on the craft skills of local engineers and workers (Myllyntaus, 1989; Nykänen, 2018), the concentration of development activities at Central Laboratory systematized these efforts. This concentration led Central Laboratory engineers to reshape learning by doing and trial-and-error by combining these practices with scientific knowledge and a systematic approach to the development of new tools and practices that could enhance work at the mills.

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Balancing Scientific Accuracy and Practical Usefulness

Central Laboratory operated at the intersection between scientific laboratory work and practical work at the pulp and paper mills. Therefore, Central Laboratory engineers had to consider the practical usefulness of their work to ensure it was applicable at the production sites. Balancing scientific accuracy and practical usefulness constitutes the second way in which elements of craft logic manifest in engineering work at Central Laboratory. It is composed of three themes: (1) relating laboratory and shopfloor practice; (2) comparing the practical usefulness of methods and equipment; and (3) observing unexpected problems that emerge from practice (see Table II for illustrative quotations).

Relating laboratory and shopfloor practice focused on recognizing how things were done in theory and practice as well as how laboratory work differed from shopfloor work. Differences between these two ways of working could then be mitigated by incorporating practical elements into laboratory work and into the methods and equipment being developed. For example, during a project to compare different methods for determining the amount of wood splinters in mechanical pulp, the following action was taken to ensure the practical usefulness of one of the methods:

> So that the standard series would not change in colour over time, it was yellowed by keeping it in a drying oven at +105 °C for 24 hours. (Mechanical pulping report 15 from 1950)

This instance depicts how the screen of a lighting table was heated overnight so that its colour would not change after heavy use in a papermill, as its luminescence would be reduced as a result. It shows how in-depth knowledge of shopfloor practice was integrated into the development processes and their outcomes. Whilst this solution was scientifically inferior, it was more practical given how work was done at the mills. Shopfloor practice also influenced what was valued in laboratory work since it shaped the ethos of work. For instance, when developing a new method to measure the strength of pulp, the following prerequisites were set:

> The most important criteria for a good method for measuring strength are the following: 1) good repeatability, 2) low human variability, 3) good fit with factory conditions, and 4 speed. (Beating and strength testing report 23 from 1960)

While the project continued with exacting scientific work to develop a new standard, the foundations of the project were laid on practical utility which the engineers interpreted as an ideal for those working on the shopfloor. In this way engineering work could oscillate between the goal of making practical and useful things and the creation of repeatability and standardization. However, the importance of understanding shopfloor practice remained central to engineering work even when methods and equipment became formalized.

Balancing accuracy and application also included *comparing the practical usefulness of methods and equipment*. These activities helped determine what constituted good work when

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Table II. Balancing scientific accuracy and practical usefulness: Themes, categories, and quotations

Second-order themes and first-order categories	*Illustrative quotations*
1. Relating laboratory and shopfloor practice	
A. Recognizing differences between theory and practice	The values that chemists and physicists use to determine the properties of technical products and substances are to a large extent clearly arbitrary. Their function is to demonstrate how the substance in question reacts to different factors, which in certain situations corresponds to those seen in practice, and therefore they enable evaluating key characteristics without complex theoretical reasoning. (Analysis and research methods report from 1942)
B. Acknowledging how laboratory work differs from the shopfloor	We found out that there was an increase in the temperature of the vat even in the end stage of the cook when external heating through the calorisator had been stopped. This had to come from – because no other explanation exists – the exothermic nature of the boiling process that can be observed with large vats but not with small ones whose cooling surfaces are proportionally much larger compared to the volume of the vat. (Cellulose experiments report 10 from 1936)
C. Incorporating practical elements into laboratory work	A method used by one of the field laboratories was taken as a starting point and some of its details were altered based on results that were received from tests to make it more suitable. (Cellulose theory report 16 from 1952)
D. Incorporating practical considerations into new methods and equipment	The CL method (A2111), which gives results already after one day, does not have the pre-cooking phase of alkali cellulose. The consequence of this is that the obtained X-viscosity values do not matter to the shopfloor staff. (Cellulose theory report 20 from 1953)
2. Comparing the practical usefulness of methods and equipment	
E. Comparing the practical usefulness of different methods and equipment	The CL colorimeters have again been compared to each other and two recently acquired machines were also studied. (Cellulose analysis report 19 from 1952)
F. Determining the most practical method or equipment for a purpose	Gurley defends its place as board tester in quality monitoring above all due to its routinized nature, but the stiffness values of thinner papers were observed to be too small compared to board values in case the EI-values given by the Steenberg pull device are used as the measure of stiffness. (Beating and strength testing report 26 from 1962)
3. Observing unexpected problems that emerge from practice	
G. Identifying specific problems related to the use of methods and equipment	The coefficients that have been in use up until this point have been developed at the same time as the method in the year 1934. Because pulp qualities have changed quite a bit after this, it was deemed necessary to revise the accuracy of the coefficients (Fibre analysis report 2 from 1958)
H. Identifying general problems related to shopfloor practice	Problems related to fluffing have become very topical in recent years especially since the usage area of offset printing has broadened. (Paper analysis report 196 from 1967)

performing a task and involved determining the best way to carry out a task. For instance, in successive projects to determine which machine was best for sorting wood fibres, the following was noted about a new device:

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> To determine the sorting capacity of the Bauer-McNett fibre sorter, sorting experiments have been done. The device has been compared with the Clark-Brainerd sorter of Central Laboratory. The results show that the Bauer-McNett sorts fibres better than the Clark-Brainerd. (Mechanical pulping report 14 from 1950)

Comparing the new Bauer-McNett device and the older Clark-Brainerd led to a realization that the new device rectified known deficiencies in how the older device worked. In addition to efficiency, these comparisons often emphasized that some methods were 'more pleasant to use' or 'simple and effortless' which shows how Central Laboratory engineers reinterpreted the ethos of work at the mills. The lack of practical usefulness also made some superior methods infeasible, even when they were more accurate than other options. For instance, when studying the water absorbing capacity of tissue papers, the lack of practical usefulness influenced what methods would be adopted:

> Of the studied methods, only submersion test gives a clear picture of absorption level, but this method was noted to be too complicated for practical use. Other methods primarily measure absorption speed, where water droplet test was deemed to be a good general method for all different tissue qualities […] The Klemm method separates one ply tissues quite well from each other and is practical, but it does not work for two ply tissues because the layers separate too easily. The working group therefore suggests that, in addition to water droplet test, Klemm method would be used for one ply tissues, and the incline plane method would be used for tissues with multiple layers. (Paper analysis report 259 from 1970)

As we can see here, the practical usefulness was a major concern that impeded the use of submersion test, which was the most accurate method, while water droplet test and Klemm method were deemed to be practical and accurate for specific purposes. Therefore, while the Central Laboratory engineers pushed the boundaries of knowledge by developing new methods and equipment, sometimes they explicitly recognized the unattainability of a uniform, standardized method and had to accept the multiplicity of ways of doing things depending on specific context. In doing so, the work to compare different methods enabled defining the best available method to use at a given time.

Balancing scientific accuracy and practical application also enabled the use of deep knowledge of practice to *observe unexpected problems that emerge from practice*. This helped in understanding issues related to using different methods and equipment, as well as the broader problems related to work at the paper mills. For instance, as the use of plastics increased in the pulp and paper industry, their properties needed to be studied:

> Different kinds of plastics are being increasingly used in the cellulose industry and especially in the bleaching facilities. Corrosion committee has therefore concluded that it is important to develop laboratory methods which can be used to study the durability features of new and existing plastic compounds. (Inorganic chemistry report 13 from 1962)

As can be seen from this example, the increased use of plastics generated a need to study their durability since corrosion was an issue that many facilities faced. In this way the

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adoption of new materials and methods opened new research areas both in terms of the materials and methods themselves but also through the development of methods with which such properties could be studied. In doing so, practical needs shaped which problems were addressed at the boundaries of existing knowledge, thereby guiding engineering work.

To theorize this dimension, balancing scientific accuracy and practical usefulness helps to align correctness and practicality (Sennett, 2008) since practical usefulness is a defining feature of the work that follows craft logic (Becker, 1978; Kroezen et al., 2021). In contrast to transforming ideas into useful objects that focus on pushing the boundaries of knowledge related to materials and objects through trial-and-error, balancing scientific accuracy and practical usefulness focuses on understanding the practices and values related to shopfloor work and reinterpreting what they mean in laboratory work. These balancing activities helped utilize practical expertise to make abstract and formal knowledge more useful (Barley, 1996; Kroezen et al., 2021) and leverage past knowledge as a springboard and testing ground for new ideas (Yamauchi and Hjorth, 2024). They show how Central Laboratory engineers adopted and reinterpreted the ethos of practicality from work in the pulp and paper mills to guide their own work. This approach helped mitigate challenges that might have arisen from strict adherence to scientific correctness without consideration of practical usefulness.

Building Engineering Community and Identity

Central Laboratory had a distinct role in the Finnish pulp and paper industry. Its function was to develop new methods and equipment, advise pulp and paper mills on their implementation, and respond to the needs of the Finnish engineering community. In addition to this national role, Central Laboratory interacted with foreign laboratories, mills, and equipment producers to stay at the forefront of international development. Based on this function, the third way in which elements of craft logic manifest in engineering work at Central Laboratory is through the building of an engineering community and a professional identity. It includes four themes: (1) establishing and updating best practices; (2) integrating community knowledge and skills into projects; (3) recognizing problems in the engineering community; and (4) claiming quality (see Table III for illustrative examples).

A key function of Central Laboratory was to *establish and update best practices* at the Finnish pulp and paper mills. For this reason, Central Laboratory engineers documented best practices, updated existing practices, and harmonized domestic and foreign practices. They also produced detailed documentation intended to influence existing practice. For instance:

> The attached report first gives a narrow review of literature on acetylation and the performed inspection tests and in the end it presents detailed instructions on how to perform acetylation. (Cellulose theory report 16 from 1963)

As we can see here, the projects not only created new knowledge on acetylation but this knowledge also led to the presentation of best practice on how acetylation should be performed. Over time, this work generated an evolving series of best practices, which was documented in the 'Testing and Analysis Methods' series and distributed to production sites. These documents influenced best practice in the Finnish paper engineer community by setting standards for good work (Sennett, 2008). In doing so, advances

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Table III. Building engineering community and identity: Themes, categories, and quotations

Second-order themes and first-order categories	*Illustrative quotations*
1. Establishing and updating best practice	
A. Establishing and documenting best practices	Standardizing the sampling methods could also be recommended. Then the samples could be cut to the same length (e.g., about 100 g), weighted with the same precision (+ − 0.2 g) with Lindell scales, and drying could be done at a specified temperature (105 °C) in two or more standardized small drying cabinets. (Cellulose analysis report 2 from 1948)
B. Updating existing practices	The aforementioned quick testing method can be considered only for the continuous quick checking of production, but it cannot be considered to be as reliable as CL method A1011, so it cannot replace it as a method for determining the strength of cellulose. (Beating and strength testing report 17 from 1955)
C. Harmonizing domestic and foreign practices	In the experiments performed at Central Laboratory, nothing has come up that would speak against using the Cross-Bovan method if used in its Swedish CCA 7 form. Taking these aspects into account, we can propose this method also as a standard method in Finland. (Cellulose analysis report 1 from 1948)
2. Integrating community knowledge and skills into projects	
D. Using knowledge from the engineering community	According to literature, testing plastics has involved measuring different characteristics related to mechanical strength and electric conductivity, in addition to measuring the aforementioned changes in weight, length and thickness. (Inorganic chemistry report 13 from 1962)
E. Collaborating with industry partners	To determine how crooked the fibres are, experiments were jointly done with the American Curlator Corporation, who, following our request, curled batches of Rosenlew's black and blue mass. (Cellulose chemistry report 30 from 1957)
F. Collaborating with other laboratories	Frag tests on 15 paper grades were performed not only at Central Laboratory but also at a Swedish and a German laboratory. (Beating and strength testing report 24 from 1961)
3. Recognizing problems in the engineering community	
G. Recognizing problems in existing knowledge developed in the engineering community	From the literature, we have not found any proper methods to objectively measure reddening. Until the end of last year, a lot has been written about the causes of reddening and how to prevent it, but as we have noted, different studies have been limited to approximately comparing different pulps after they have been treated with saline solution, hydrogen peroxide, sodium nitrite and other solutions. (Cellulose analysis report 4 from 1949)
H. Responding to problems coming from production sites	At the 18.3.54 meeting of the analysis sub-committee, engineer Klingstedt proposed that Central Laboratory would develop a quick method for the strength determination of cellulose. (Beating and strength testing report 17 from 1955)

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Table III. (Continued)

Second-order themes and first-order categories	*Illustrative quotations*
4. Claiming quality	
I. Associating work quality with third parties	In the comparison between the analysis results of Central Laboratory and seven foreign laboratories, it has been shown that the values arrived at by Central Laboratory are close to the values of the majority of other laboratories. (Mechanical pulping report 31 from 1955)
J. Declaring the quality of work to third parties	If instructions are followed and work is carried out meticulously, the specification can be done with satisfactory precision. (Cellulose analysis report 14 from 1951)
K. Taking individual credit for completed work	The research in question, except for the thesis project, focused on analysing the influence of tar, which was supervised by DSc H. Sintola and led by MSc (Eng) Paul Ålander, and the last mentioned also prepared the project report. (Cellulose analysis report 10 from 1951)

that pushed the boundaries of established knowledge were translated into best practices, enabling mills to replace local practices and ad hoc problem-solving with more systematic ways of working.

Work within the engineering community also included *integrating community knowledge and skills into projects*. In practice, Central Laboratory engineers incorporated knowledge from engineering reports and professional journals into projects and collaborated with paper mills, industry partners, and other laboratories. For instance, when studying the use of the light microscope to analyse paper surface, the project started with a literature review with the following conclusions:

> Researchers have a rather uniform opinion of the benefits of the shading method. As we already mentioned at the beginning of the report, direct shading can be used only for thin and translucent paper. (Fibre analysis report 3 from 1965)

Using this knowledge helped to establish what was known about the subject and set a starting point for the project. In doing so, engineers could use existing knowledge to determine the boundaries of what was known and direct their work to focus on the cutting edge of development.

Being part of the engineering community also enabled *recognizing problems in the engineering community*. These activities included pinpointing shortcomings in existing knowledge and responding to issues coming from pulp and paper mills. Many projects originated directly from industry-wide technical committees that requested studies on various aspects of pulping and papermaking. This interaction influenced Central Laboratory in determining what topics were studied but also helped to build an understanding of why this work was needed in the first place. By recognizing problems within the engineering community and integrating community knowledge and skills into its projects, Central

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Laboratory functioned not in isolation but as part of the broader industry. This embeddedness facilitated the adoption and reshaping of practices originating in pulp and paper mills while also fostering an understanding of the prevailing ethos of practicality.

While the community interactions focused on setting standards and advancing knowledge by recognizing what was already known and where potential problems might lie, it was also about *claiming quality* of the work accomplished at Central Laboratory. A key dimension of this work was to follow international standards and signal work quality to third parties. For instance, when comparing different methods to bleach pulp, the following was noted:

> When comparing the results achieved by following the Swedish and Finnish standard methods (a and e), we can observe that their congruence is very good. (Cellulose analysis report 21 from 1953)

This finding indicated that the work done at Central Laboratory did not deviate from the results of foreign laboratories. These claims of quality work were communicated not only to other laboratories but also to third parties such as prospective customers. Furthermore, claiming quality work also extended to individual engineers, as new devices could be named after the inventor, and engineering reports were prepared and signed by the chief engineer in charge of the project. This work created a distinction between engineers who had the skills to develop new methods and equipment at their discretion and those whose job was to use them.

This dimension reflects the communal aspects of craft logic (Kroezen et al., 2021; Kroezen and Heugens, 2019; Raviola et al., 2023; Sennett, 2008) since Central Laboratory had a key role in developing the Finnish paper engineer community. These activities helped Central Laboratory engineers access distributed knowledge within the engineering community (Barley, 1996) and define quality standards (Sennett, 2008). Simultaneously, they helped to construct an identity for Central Laboratory and individual engineers working there as industry forerunners (Bell et al., 2018) and communicate this commitment to quality to customers and third parties (Cattani et al., 2017; Holt and Yamauchi, 2019). These activities created a distinction between Central Laboratory engineers who applied their craft to the development of new methods and equipment at the boundaries of existing knowledge and engineers at the pulp and paper mills whose job was to apply that knowledge when it was translated into best practices and standards.

How Craft Influences Engineering Processes and their Outcomes

To understand how craft influenced engineering processes, we reanalysed the projects using our coding scheme. Comparing crafting activities across projects revealed that each aggregate dimension involved second-order themes related to problem-finding and problem-solving. These activities occurred cyclically, and the resulting solutions influenced work outcomes as they became embedded in projects (see Figure 3). In the following, we analyse these cycles in greater detail, while Table IV presents illustrative evidence of both the cycles and their influence on work outcomes.

Problem-finding activities originated from requests by technical committees or unexpected issues concerning methods, equipment, or practices. Community-originated

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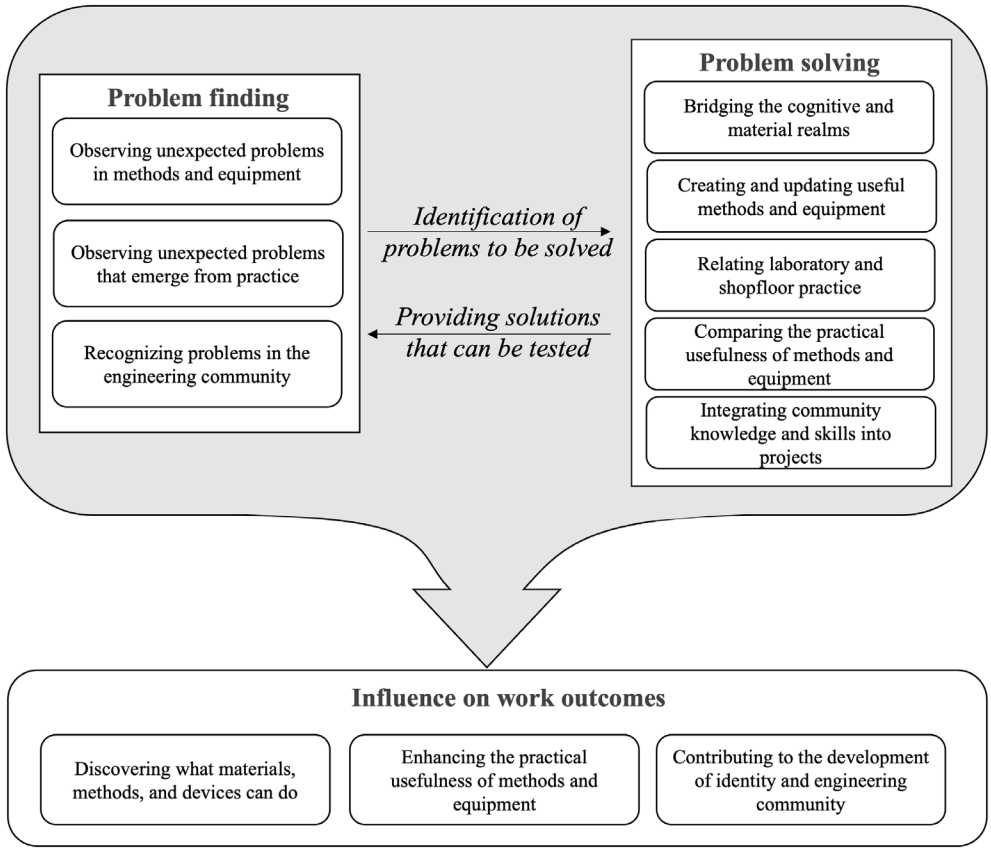


Figure 3. Interactions between problem-finding, problem-solving and work outcomes

problems often catalysed project initiation, while unexpected issues were related to the developed solutions. For example, a project to measure and compare the hardness of wood types began from a committee request:

> The goal of the current project, which has been part of the analysis committee programme since 7.2.1947, is to establish a basis following the suggestion from the chemist meeting (11.10.1946) to select a hardness number for determining the overall evaluation of the hardness of unbleached cellulose. (Cellulose analysis report 8 from 1949)

While solving this issue was successful, it revealed a new problem related to other wood types:

> It is still worth noting that the now generated conversion functions pertain only to the hardness figures generated by normal fir sulphites and pine sulphates. If the pulp is produced from other tree species, we do not yet know the relationships between the figures. (Cellulose analysis report 8 from 1949)

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Table IV. Illustrative examples of problem-solving cycles and their influence on work outcomes

Problem-finding activities	*Problem-solving activities*	*Influence on work outcomes*
Following the decision made by the analysis sub-committee on 4.11.48, CL has compared a few standard methods for determining the chemical and physical constants at the same time when studying other characteristics is ongoing.	The relationship between figures determining alkali solubility and gum content is good for some figures but on the contrary bad for some figures.	It was still considered that the Halsen method is still "the most pleasant" to perform, although the human factor is more significant than in other methods. (Cellulose analysis report 9 from 1950)
Boiling tests whose goal is to show what kind of similarity can be attained, especially regarding quality, when comparing boiling at the factory and boiling on a laboratory scale.	Replication of Sunila's boiling curve in the laboratory demonstrated that pulp prepared in the laboratory was significantly rawer compared to boiling at the factory.	From the performed experiments and measurement, we deem that pulp quality, if it pertains to the qualities of strong sulphite, shows good consistency between values received from boiling in a factory and using the same method for boiling in a laboratory. (Chemical pulping report 10 from 1939)
Analysis subcommittee meeting on 25.10.51 incorporated studying Vereinigte Glanzstoff-Fabriken AG (VGF) methods into the research programme.	One method intended to be the German standard for determining the common alpha number and the quantity of cellulose that does not dissolve in different lye strengths (by diluting lye after mercerizing) has been compared to equivalent Swedish methods.	The alpha number method does not seem to be more beneficial than the well-integrated Swedish alpha number method. On the contrary, it seems that the proportion that does not dissolve in alkali can be determined more surely with the German method, which, in its standard form, cannot be directly recommended for routine use due to its slowness and laboriousness. (Cellulose analysis report 22 from 1953)
Laboratory experiments have proven that studying different pulp qualities with Jayme's method can generate great difficulties due to the colloidity of different solutions, especially when it comes to centrifugation and gravimetric determination.	For this reason, there was a reason to change this method slightly to suit our circumstance. Acetylation is performed without activation and the measuring of results has been simplified so that directly after acetylation, an optic opacity measurement is made from the reaction solution with a Lumetron-photocolorimeter.	The performed experiments have demonstrated that the method gives enough precise and repeatable results, which acetylation experiments rarely produce. (Rayon report 3 from 1948)

(Continues)

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Table IV. (Continued)

Problem-finding activities	*Problem-solving activities*	*Influence on work outcomes*
As already mentioned, three pulp grades were ground with a Lampen mill to R-S grades of 20, 40 and 60 so that we could determine how to reduce the drying times of sheets without it affecting results.	These show that drying sheets with the CL drum with two or even one absorbent carton gives the same strength values for bleached sulphite; unbleached sulphites generate approximately the same values, and unbleached sulphates at the beginning of grinding also give very similar values in terms of all strength measures.	The aforementioned quick method can be considered as a method that a mill can use in rapidly measuring production, but it cannot be considered as reliable as CL method A1011, and it cannot replace this method for measuring pulp strength. (Beating and strength testing report 17 from 1955)
In addition to this, as mentioned, following the request of the analysis committee, the necessity to use nitrogen had to be studied.	The findings show that the weight of the bullets increase as they are repeatedly used, which generates values that are too big. The difference is approximately 2 per cent for each 0.1 mg – not evenly distributed – which results from the alteration of the size and surface area of the bullets.	If precise results are needed, each fall requires an unused bullet. (Cellulose analysis report 3 from 1949)

In this way problem-solving activities followed problem identification and focused on creating solutions that successfully materialized ideas, were useful to the industry, and contributed to the engineering community. Technical and practical considerations were often intertwined when solving a problem, as translating ideas into objects required considering their practical usefulness. For instance, the development of a wood consumption meter started with updating the design of an existing machine:

> At first, this problem was solved by installing a pressure gauge to each press. The impulses flowed through the pressure gauge, which was configured so that the connection between the impulse giver and the plotter was disconnected when the pressure in the cylinder was low. (Mechanical pulping report 35 from 1957)

However, the new solution quickly revealed that shopfloor work was less sophisticated than laboratory work, and the solution proved impractical.

> However, this construction was not simple enough or cost effective […] So, this construction was abandoned, and another was created, which did not require a pressure gauge. (Mechanical pulping report 35 from 1957)

As can be seen here, the newly developed solution fed back into problem-finding, as the solution was not practical enough when tested, which then informed problem-solving. What these problem-solving activities highlight is that instead of concentrating

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on a single dimension at a time, problem-solving cycles oscillated between the material, practical, and social dimensions that are held together by the craft logic.

The cyclical problem-solving activities influenced work outcomes when they helped reveal what materials, methods, and devices could do, enhanced the practical usefulness of methods and equipment, and contributed to the development of identity and the local engineering community. The project focused on the development of a wood consumption meter realized these outcomes by providing a new technical solution that was both practical and provided clear technical benefits. Conversely, the project focused on the hardness of wood types and concluded that while there was a precise way to measure the hardness of wood, the device used for these measurements was 'quite fragile' and 'only very skilled engineers could flawlessly operate the device'. Therefore, while the project pushed forward understanding on how to measure the hardness of wood, the measure could not be standardized because it required highly skilled craftspeople to make the measurements. Thus, the project could not displace craft with a standard, and the project report finished with a remark that the development of a standard measure would require a new method.

Problem-solving cycles are typical in engineering (Brusoni, 2005; Nersessian, 2006), but they also characterize craftwork (Bozkurt and Lara Cohen, 2019; Sennett, 2008). These cycles facilitate the integration of existing knowledge and techniques to achieve high-quality outcomes while encouraging experimentation and discovery in the search for new solutions to the problem at hand (Fox Miller, 2017; Holt and Popp, 2016). As our findings show, the ability to combine the material, practical, and social dimensions demonstrates all-roundedness (Kroezen et al., 2021), as engineers combined multiple techniques to create new solutions. This required blending abstract, embodied, and distributed knowledge (Bettiol and Micelli, 2014; Bozkurt and Lara Cohen, 2019), which extends beyond the exacting and disenchanting work of science (Suddaby et al., 2017). Thus, craft was not merely an epiphenomenon of engineering processes but an essential component of engineering work that contributes to the final design, much like its other dimensions of engineering, such as science and systems thinking (Petroski, 1996).

DISCUSSION AND CONCLUSION

The goal of this study was to advance our understanding of how and why craft enables industrialization. To answer this question, we analysed engineering work at Central Laboratory, which developed methods and equipment for Finnish pulp and paper companies and played a crucial role in the industrialization of pulp and paper production in Finland. Our findings demonstrated how elements of craft logic were integrated into engineering work and processes at Central Laboratory. We now return to the theoretical foundations of this study to discuss how our findings contribute to understanding the role of craft in industrialization.

The Plasticity of Craft Logic and how it Contributes to Industrialization

Our findings reveal how elements of craft logic are reflected in the way Central Laboratory engineers transformed ideas into useful objects, struck a balance between scientific accuracy and practical usefulness, and fostered a sense of community and identity. These dimensions show how the practices, skills, and attitudes associated with craft, originally

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embodied in ad hoc measurement practices and mill-specific ways of managing the production process, were adopted, reinterpreted, and reshaped by Central Laboratory engineers to develop uniform methods and equipment for the pulp and paper industry. We theorize that this results from the *plasticity of craft logic*, which allows it to be transferred into different contexts and reinterpreted to address new objectives.

Plasticity relates to craft as an institutional logic. Prior research on the craft logic has either assumed a linear evolution from craft to industry (Stinchcombe, 1959; Thornton, 2002) or highlighted how craft logic can be regenerated to create innovation and change after widespread industrialization (Kroezen and Heugens, 2019; Rennstam and Paulsson, 2025). Yet, institutional logics are also characterized by plasticity, which allows individuals to reshape and reinterpret logics to meet new goals or apply them to new contexts (Notteboom et al., 2013; Strambach, 2010). In doing so, plasticity relates to stretching practices, attitudes, and roles to new forms without breaking the logic (Ghaffari et al., 2022; Lok and De Rond, 2013).

Plasticity conceptualizes the adoption and reinterpretation of craft logic in the work performed at Central Laboratory, as the industry scaled up to mass production. During this process, the engineering work focused on developing new methods and equipment to address an evolving set of problems. Since the development work had limited repeatability and initially minimal formal knowledge and standards, engineers relied on craft practices, skills, and attitudes to determine what various materials and techniques could accomplish and how they should be applied to produce the best possible work. In doing so, handwork and craft-based control over technologies were employed to enhance machine-based manufacturing (Holt, 2026). When the basis for repetitive work was missing, as is often the case when creating new things, there was ample space to approach work as a craft with the goal of developing new and useful things. As a result, the craft logic was reshaped to meet this need as Central Laboratory engineers reinterpreted its meaning in a new context.

The plasticity of craft implies malleability that results in durable change as actors both shape craft into a new form and become influenced by these changes (Giddens, 1984). While much of the craft related to pulping and papermaking disappeared from individual mills due to the standardization of processes, the practices, skills, and attitudes associated with craftwork endured at Central Laboratory. Therein craft transformed into *preparatory workmanship*, which enabled mass production by developing its plants, tools, and methods (Pye, 1995, pp. 131–132). Thus, instead of viewing craft and industrial logics as opposites (Ravasi et al., 2025), these two logics coexisted, as craft skills and attitudes guided the development of the preconditions for standardized industrial production (see also Holt and Popp, 2016; Pye, 1995). Therefore, plasticity not only sustains the status quo through temporary change (Lok and De Rond, 2013) but can also lead to more enduring change (see also Ghaffari et al., 2022), as is the case with craft.

Craft Frontier at the Forefront of Industrialization and Standardization of Work

The literature on craft presents craftwork and standardized work as opposite ends of a continuum (Kroezen et al., 2021). It also presents craft as something that disappears

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during industrialization because of an increasingly fragmented division of labour, control mechanisms, and the deskilling of workers (Braverman, 1974). Our work contributes to this literature by arguing that craftwork can facilitate the emergence of standardized work, which is a key component of industrialization.

On the one hand, standardized ways of working are created through craftwork, as we have already argued. Developing a new standard requires understanding how ideas can be executed in practice and how they should be applied to ensure practicality. Industrialization and standardization, therefore, require craftwork to find ways to standardize handwork (see also Holt and Popp, 2016). Standardization also requires communal ways of working, as many people apply standards. Even though standardized processes may be imposed upon workers, the standards are established by committees of experts. The skills and attitudes associated with craft can therefore contribute to standardization when standards for good work are codified and broadly applied. Braverman's (1974) seminal work is surprisingly absent of the craft that goes into developing production systems and control mechanisms.

On the other hand, standardizing work opens avenues for further exploration through craft. When a reliable method for measuring a quantity exists, resources can be directed to finding ways to improve that quantity in a desirable direction. It also frees up resources to explore other characteristics that may be relevant for the quality of the outcome. Keeping some things constant makes it possible to explore other characteristics more deeply and widely. Over time, this work may necessitate revisiting and recrafting previously agreed-upon standards as the application area evolves and the criteria for good work change.

To generalize these dynamics, we suggest that there is a *craft frontier* that perpetually remains beyond standardization, as craftwork is essential for creating standardized work, which then opens opportunities to establish new standards. This frontier is inherently characterized by uncertainty that creates opportunities and invites *preparatory workmanship*, which then produces the preconditions for mass production (Pye, 1995, pp. 131–132). At the craft frontier, individual judgment, the mastery of techniques, and an explorative mindset are used to develop new candidates for future standardized ways of working. In doing so, craftwork precedes standardization and creates conditions for it since craftspeople possess knowledge that enables them to control technology (Dormer, 1997, p. 102).

The craft frontier is an epistemic boundary that constantly shifts as craftwork generates new standardized ways of working, and the focus shifts to the next problem to be defined and solved. The craft frontier is controlled by the engineering community, which determines its direction and sets the criteria for good work in accordance with the industrial logic and its ideals of standardization and efficiency. The engineers operating on the frontier have considerable discretion and can approach their work as a craft, but their proposals may be overruled by the engineering community. Therefore, the engineers working on the frontier and the wider community of engineers co-construct the craft frontier through a communal process which can alleviate conflicts (Besharov and Smith, 2014) and help to control the process of industrialization (Holt, 2026). In consequence, the craft frontier allows the craft and industrial logics to co-exist and the

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engineers following different logics to work toward the common goal of industrialization (Reay and Hinings, 2009).

Engineers operating inside the frontier follow industrial logic that standardizes and organizes their work. On the other side of the frontier lie non-systematic and ad hoc ways of working as well as natural phenomena that are not yet well enough understood to allow the development of systematic, standardized ways of doing things. As the craft frontier moves forward, an increasingly large portion of this reality is organized to function according to the industrial logic. However, industrial logic does not subsume craft at the frontier because there is always more of reality to work with. Engineers work with natural phenomena and nature always offers more to tinker with. This is not a question of limited natural resources, but of the physical and chemical phenomena that are omnipresent and where our understanding remains limited. As the craft frontier moves forward epistemically, there are always more opportunities for craft. This means that craft logic continues to play an important role even though industrial logic organizes an increasingly large portion of social reality.

The craft frontier also has implications for craft politics because it creates inequalities within the engineering community. Previous discussions of craft politics have primarily focused on reclaiming history and agency through craft (e.g., Gasparin and Neyland, 2022; Vachhani et al., 2025). However, Layton's (1971, p. 57) theorization of the societal role of engineers highlights their responsibility to improve society and ensure that technological progress benefits humanity. The craftwork that enables such improvements simultaneously creates inequalities among engineers. While some engineers are able to hone their skills and experiment with new ideas at the craft frontier, others are left to follow the standardized ways of working and relinquish many of the craft elements of their work at the mills. Importantly, this development is not contested because engineers share a professional understanding that standardization is necessary for societal progress. As a result, the inequalities created through craft emerge as a consequence of engineers' professional ethics and commitment to improvement. This argument complements Braverman's (1974) observation that engineers reduce the discretion of other occupational groups through standardization by demonstrating that standardization also creates inequalities within the engineering occupation itself.

Overall, our findings provide a more nuanced understanding of the role of craft in industrialization. Rather than being displaced by industrialization, we have shown how craft logic contributed to the emergence of mass production and the broader societal change it enabled. Our findings position craft logic as a source of change because it contributes to the creation of new knowledge, technologies, and practices that can be subsequently standardized and scaled for mass production. At the same time, change may also arise from the plasticity of craft logic. As elements of craft logic are reinterpreted to address new objectives, they can give rise to novel practices and organizational arrangements. Taken together, our study shows that craft logic can contribute to organizational and societal change in two ways: by generating new tools, technologies, and practices and by providing a foundation for novel forms of organizing through its reinterpretation in new contexts. We hope these findings encourage management and organization scholars to examine how craft contributes to change across different organizational contexts while remaining a distinct approach to work and organizing.

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Limitations and Suggestions for Further Research

This study has some limitations that can inform future research. First, our study relies on historical analysis to theorize about the plasticity of craft. Although this approach aligns well with our goal of examining how craft enables industrialization, we acknowledge the need to explore the plasticity of craft further. By studying how elements of craft are adopted in different lines of work, we can better understand which elements can be more easily adopted to fit various organizational contexts. Simultaneously, studying the plasticity of craft can further our understanding of the different functions that craft can have in societal transformations.

Second, our study relies on data from a single organization to develop the idea of the craft frontier. While Central Laboratory was responsible for advancing the Finnish pulp and paper industry, examining other sectors could enhance our understanding of how the craft frontier operates. This could involve observing the dynamics of the craft frontier in real time using ethnographic methods to understand how skills and attitudes related to work differ between the frontier and standardized work. It would also be worthwhile to study the micropolitics of how the craft frontier becomes differentiated from standardized work and how this distinction is maintained.

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